



December 2025

(For CS)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

20 Marks

- 1-

9. What is the output of the following code?
`s = "Python"`
`print (s [: 3])`
 A) P B) Py
 C) Pyt D) hon
10. What will happen if you try to modify an element in a tuple?
 A) The element will be updated B) The element will be ignored
 C) The tuple will be converted to a list D) Error will be raised
11. Which operator is used to concatenate two strings in Python?
 A) + B) &
 C) # D) *
12. How do you insert an element 100 at the 3rd position in the list, num = [10, 20, 30, 40] ?
 A) num.append (100) B) num [2] = 100
 C) num.insert (2, 100) D) num.extend (100)
13. What does an object consist of in Python?
 A) Only attributes B) Only methods
 C) Attributes and Methods D) Parameters and Arguments
14. What does the 'r' mode do when opening a file?
 A) Write to the file B) Read from the file
 C) Append to the file D) Delete the file
15. What happens if you open an existing file in 'w' mode?
 A) It appends new data to the file B) It raises an error
 C) It renames the file D) It overwrites the file content
16. What does the acronym CSV stand for in the context of Python files?
 A) Comma Separated Values B) Compressed System Versions
 C) Character String Variables D) Custom Structured Vectors
17. What is the result of the following code?
`d = { 1 : 'red', 2 : 'blue'}`
`print ( d [3] )`
 A) IndexError B) ValueError
 C) KeyError D) TypeError
18. How many errors does the following code contain?
`for i in range(5)`
`print (i)`
 A) 1 B) 2
 C) 3 D) 4
19. What is the typical file format used for exporting data from Pandas to a file?
 A) JSON B) HTML
 C) XML D) CSV
20. What method is used to display a plot created with Matplotlib?
 A) plt.show() B) plt.plot()
 C) plt.display() D) plt.render()

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the features of python programming language.	8	L*1	1	1
	b) Write a python program to check if a given year is a leap year or not.	8	L3	2	2
2.	a) Explain conditional and iterative statements in python with examples.	8	L2	1	1
	b) Define module and explain the different ways to import module in python.	8	L1	2	2
3.	a) Mention the rules to be followed in the selection of variables in python and classify the following into valid and invalid identifiers: (i) num_2 (ii) +add (iii) _name (iv) email@123 (v) 1_value (vi) roll.no	8	L1	1	1

- b) Define function and write its syntax. Explain the types of functions. 8 L2 2 2

Unit – II

4. a) Define list in python and explain creating, accessing and traversal of a list with examples. 8 L1 3 1
- b) Write a python program to copy the content of one text file to another text file. 8 L3 4 4
5. a) Consider a tuple, x = ('one', 'two', [3, 4, 5], '6')
Write the output for the following statements:
(i) print (x [: :])
(ii) print (x [1] [: : -1])
(iii) print (x [2] [: : -1])
(iv) print (x [-4] [0]) 8 L2 3 1
- b) Explain the various ways to write data on to a file in python with an example to demonstrate each method. 8 L2 4 2
6. a) Define dictionary in python..Explain items(), keys() and values() methods in dictionary with examples. 8 L1 3 1
- b) Describe the relation between a class and an object in Object-Oriented concept. 8 L2 4 2

Unit – III

7. a) Rewrite the following codes by removing the error(s) (if any) to get the desired output. Write the output for the corrected codes.
(i) x, y = 10, 20
Sum = X + Y
Print("The sum of numbers is equal to" sum)
- (ii) While i < 10 ;
Print (i, end = " ")
i = i + 1
- (iii) m = input ('Enter your marks: ")
If marks => 40
print ("You have passed")
or
print ("You have failed")
- (iv) for i in range(1, 6) :
print("6 x ", i, "=", 6 * i) 8 L1 5 3
- b) Write a python program to create a pie chart for student strength shown in the table:
- | Dept. | Civil | CS | BT | Mech | EE |
|------------------|-------|-----|----|------|-----|
| Student Strength | 49 | 316 | 66 | 128 | 100 |
- Add the title, legend and label to the pie chart and highlight the student strength of Civil department by detaching it from the pie chart. 8 L2 5 3
8. a) Explain DataFrames and Series in Pandas. 8 L1 5 4
- b) Write a python program to create a line graph for the percentage of marks scored by three students in different year as shown below:

Student Name	1 st year	2 nd year	3 rd year	4 th year
Rahul	95	72	60	92
Priya	60	98	72	98
Sachin	82	44	84	78

Add the title, axes, legend, and marker for the graph. 8 L3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
